Wearable Smart Insole for early Diabetic Foot Ulcer Prediction using Multi-Modal Sensing



Final Year Project Report

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A Dissertation
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by

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LAHORE, PAKISTAN

Department of Electrical Engineering
Information Technology University of the Punjab
Lahore, Pakistan
Year

CERTIFICATE OF ORIGINALITY

We hereby declare that this report titled “Project Title” is our own work to the best of our knowledge. It contains no materials previously published or written by another person, nor material which to a substantial extent has been accepted for the award of any degree or diploma at ITU or any other education institute, except where due acknowledgment, is made in the thesis. Any contribution made to the research by others, with whom we have worked at ITU or elsewhere, is explicitly acknowledged in the thesis.

We also declare that the intellectual content of this report is the product of my own work, except to the extent that assistance from others in the project’s design and conception or in style, presentation and linguistic is acknowledged. We also verified the originality of contents through plagiarism software.



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The undersigned hereby certify that they have read and recommended the report entitled "***Smart System for Early Detection of Diabetic Foot Ulcers***" by **Abdul Rahman (BSEE21080) and Muhammad Nouman (BSEE21063)** for the degree of Bachelor of Science in Electrical Engineering.



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DEDICATION

We dedicate this project to our families, who have always been our biggest support throughout our studies. We are very thankful to our parents for their love, sacrifices, and constant belief in us, which pushed us to aim higher and work harder. Their financial and emotional help kept us focused on our goals, even when we faced challenges. The values they taught us, like patience and resilience, are part of this work.

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Lastly, we want to thank our friends and classmates for their support during this journey. Their teamwork, friendship, and shared interest in learning made this experience enjoyable and memorable. They helped us during tough times and celebrated with us in our achievements. This project is dedicated to everyone who played a part in our growth.

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Abstract

Diabetic foot ulcers (DFUs) are a common problem for people with diabetes. If they are not found early, they can cause serious issues like infections, amputations, or even death. Most current methods to check for DFUs rely on visual checks or treatments after the problem has already started, which are often too late.

This project aims to build a smart system that can detect DFUs early using sensors, artificial intelligence, and deep learning. The system is a wearable insole with sensors that measure pressure, temperature, and moisture in the foot. The data from these sensors will be studied by an AI program that can spot early warning signs of ulcers before they are visible.

The idea behind this work is to move from late treatments to early prevention, which can reduce the load on hospitals and improve the lives of diabetic patients. By checking the feet in real time and giving predictions, the system can send alerts to both patients and doctors so action can be taken early.

Some of the main benefits of this project include fewer amputations, lower treatment costs, and better health results because of constant and simple monitoring. This type of system can be especially useful in countries where healthcare is limited, as it provides a cheap and practical solution for dealing with diabetes problems.

CHAPTER 1

INTRODUCTION

Diabetic foot ulcers (DFUs) are a big concern for healthcare systems because they cause serious disability and high treatment costs. A report from the American Diabetes Association in 2019 showed that DFUs cost the U.S. healthcare system between $9 and $13 billion each year. DFUs are also one of the main reasons for lower limb amputations, which lowers quality of life and increases the risk of illness and death for people with diabetes.

Finding DFUs early can help reduce these costs and improve patient health. To do this, regular and accurate monitoring is needed. With new technology like sensors, artificial intelligence, and deep learning, it is now possible to track foot health and predict when ulcers may form so that treatment can start quickly.

The goal of this research is to develop a smart method to detect ulcers at an early stage. This will make it possible to provide timely care and improve the quality of life for diabetic patients.

1.1 Motivation

This project is inspired by the rise of diabetic foot ulcers (DFUs) among people with diabetes. The International Diabetes Federation says that about one in four diabetics will get a foot ulcer during their lifetime. DFUs can lead to amputations and a serious drop in quality of life.

The cost of treating DFUs is also very high, reaching billions every year. This creates a huge burden on healthcare systems.

Our project aims to develop an early warning system that can spot the first signs of ulcers. It combines artificial intelligence with sensor technology. This will help improve patient care and reduce treatment costs.

At present, most detection methods rely on visual checks. These are not always accurate and can delay treatment.

1.2 Challenges

Building an early detection system for diabetic foot ulcers comes with many challenges. One issue is combining different sensors, such as pressure, temperature, and moisture, into

one comfortable and wearable device. Each sensor also needs to be carefully calibrated to give accurate results.

Another challenge is collecting enough varied data to train the AI system. The dataset must include patients with different ages, health conditions, activity levels, and even different environments. This ensures the model stays reliable in real-life use.

Making the system easy to use is also very important, especially for older patients or those who are not familiar with technology. The mobile app should be simple, show alerts in real time, and avoid confusing technical terms.

There are also issues with approvals and meeting medical device standards. These steps are necessary to make sure the system is safe and can be used in healthcare.

Diabetes itself is a serious health problem that affects millions of people worldwide. Many do not even know they have it. In fact, one out of every two people with diabetes is unaware of their condition. Without treatment, diabetes can lead to heart disease, kidney damage, nerve problems, or even blindness.

The good news is that early detection can prevent many of these issues. If someone has symptoms like being thirsty often, feeling tired, or needing to urinate frequently, or if they have a family history, regular check-ups can help. Small lifestyle changes like eating healthy, exercising, and managing weight also play a big role. Early action is the best way to avoid complications and live a healthier life. Even small steps taken today can protect the future.

1.3 Problem Statement

The main problem this project addresses is the late diagnosis of diabetic foot ulcers. Delays in finding them often lead to serious issues like infections and amputations. Mostly reactive, the classic ways of monitoring foot health depend on visual inspections instead of continuous, proactive monitoring. This dearth of early diagnosis significantly impacts healthcare costs and patient results. Permitting quick therapies and reducing the ulcer risk, the project provides a smart system for real time monitoring of foot health by integrating artificial intelligence and sensor technologies.



Figure 1: Detection of Diseases

1.3.1 Unmet Needs

This Final Year Project (FYP) aims to solve the lack of effective early detection methods for diabetic foot ulcers (DFUs). In the United States alone, more than 6.3 million people suffer from DFUs. According to the American Diabetes Association, these ulcers lead to nearly 80,000 amputations every year. This figure emphasizes how vital new therapies are to give timely warnings before ulcers grow into major difficulties. Dealing with this issue will help the suggested technique dramatically lower amputations, raise patient quality of life, and lower the financial burden on healthcare systems related to sophisticated ulcer treatments.

1.3.2 Potential Market

Diabetic patients, healthcare professionals, and hospitals mostly make up the range of customers this project could serve. The CDC estimates that over 34 million Americans are diagnosed with diabetes, with a significant number vulnerable for DFUs. For a wearable monitoring device, this produces a sizable customer base. Healthcare professionals and hospitals trying to improve patient care and avoid consequences related to diabetes provide yet another big market sector. The market for diabetes management tools is expected to grow quickly; some forecasts suggest it may be worth more than $30 billion by 2025.The recommended system can access this booming industry by offering a proactive means of monitoring foot health, therefore greatly profiting patients as well as medical professionals.

1.4 Report Outlines

This study mainly seeks to define the design, development, and deployment of an intelligent system based on sensor technology and artificial intelligence for early detection

of diabetic foot ulcers. **Chapter 1** outlines the reasons behind the initiative, challenges, and problem statement. **Chapter 2** is an all-inclusive literature evaluation covering ongoing developments in diabetic foot ulcer detection technology and study. **Chapters 3 and 4** offer a thorough survey of sensor and artificial intelligence integration by means of the proposed methodology and system architecture. Along with the timeline and team member responsibilities, **Chapter 5** presents the project milestones, task distribution, and cost analysis. **Chapter 6** at last concludes the study with a summary of the results and proposals for additional research.

1.4.1 Sustainability

This project's sustainability can be judged on a number of levels, including:

1. Improved Patient Outcomes: Better health results from the emphasis the system gives to preventive care above reactive care. Early diagnosis spares patients more invasive treatments, including amputations, therefore improving their quality of life and lowering healthcare expenses. The technology will also aid to lessen long-term consequences of diabetes, including infections and hospitalizations.

2. Diabetes and its consequences: have a major influence on healthcare systems all around. By allowing early intervention and proactive monitoring, this method helps to reduce the burden on healthcare systems thereby lowering hospitalizations and demand for costly medical treatments.

3. Accessibility and Affordability: The system uses low-cost sensors along with cloud-based analysis, which makes it affordable and easy to scale. It can work in many healthcare settings, from city hospitals to remote areas where access to medical care is limited.

CHAPTER 2

LITERATURE REVIEW

2.1. Diabetic Foot Ulcers (DFUs): Prevalence and Risk Factors

Research shows that about 15% of diabetic patients will develop a foot ulcer at some point. More than 85% of diabetes-related leg amputations happen after a foot ulcer.

Several factors cause DFUs:

- **Peripheral neuropathy:** Nerve damage makes patients lose feeling in their feet, so they do not notice small injuries.
- **Ischemia:** Poor blood flow in the legs slows down the healing of wounds.
- **Infection:** Bacteria can enter untreated wounds, and if not managed, this can lead to gangrene or even sepsis.

These factors together make DFUs very dangerous, with high chances of illness and death. This shows why early detection and preventive care are so important.



2.2. REQUIREMENT FOR WEARABLE LOAD SENSING OF DFU

In recent years, there has been a big improvement in understanding how diabetic foot ulcers (DFUs) develop. This knowledge is important for creating wearable devices that can measure pressure on the foot and help assess the risks and effects of DFUs.

• **Measurement Capabilities:** Research shows that it is important to measure both pressure and shear stress on the bottom of the foot. To do this, sensors that can detect forces in multiple directions are used at each measurement point. These sensors capture both the pressure and the shear stress.

• **Sensor Arrangement and Position:** The positioning of sensors in relation to the plantar surface is a crucial element in obtaining clinically valuable measurements. Research indicates that DFUs may appear in numerous areas on the plantar surface, and these sites can be erratic as a result of offloading measures. As a result, it is sensible to place sensors throughout the entire plantar area.

• **Spatial Resolution:** The quantity of sensing elements and their corresponding dimensions are interconnected features of the measurement system. For a specific coverage area (e.g., the plantar surface), the dimensions of the sensor component determine the highest spatial resolution that can be attained. Generally, smaller sensors are favored as they allow for greater spatial resolutions. Nonetheless, incorporating numerous sensors into a measurement system presents related requirements for interface electronics, data processing, and data management.

• **Sampling Rate:** Most commercially accessible plantar pressure measurement devices function in the range of 50–100 Hz. These rates are suitable for reflecting the plantar pressure changes linked to normal walking behaviors; thus, the system's sampling frequency (e.g., taking all sensors into account) must be at least 50 Hz.

2.2. Current Technologies for DFU Detection

The most common methods of detecting DFUs include:

- **Visual Inspections**: Often conducted by clinicians or patients themselves. However, by the time ulcers become visible, significant tissue damage may have already occurred.
- **Thermal Imaging**: Measures temperature anomalies in the foot, indicating inflammation. While useful, it is generally limited to clinic settings and is often used after ulcers have already begun to form.
- **Pressure Insoles**: Used to identify high-pressure points on the foot, which could lead to ulceration. However, pressure alone is not always a sufficient indicator of ulcer risk.

2.3. Applications of AI and Deep Learning in Medical Diagnosis

AI, particularly **deep learning**, has emerged as a powerful tool in medical diagnosis, especially in analyzing complex datasets such as medical images, sensor data, and patient health records. **Convolutional Neural Networks (CNNs)**, which are specifically designed to handle spatial data, have been successfully applied in **medical image classification** for

tasks like cancer detection, retinal image analysis, and skin lesion classification.In the context of DFUs, CNNs can process **multimodal sensor data** (pressure, temperature, and moisture readings) to detect the subtle patterns that precede ulcer formation. By training the AI model with labeled datasets, where sensor readings from diabetic patients are correlated with foot health outcomes, the system can learn to predict the early stages of ulceration.

2.4. Gaps in Existing Solutions for DFU Detection

The majority of existing therapies are reactive, which means they discover ulcers after severe harm has already happened. The main deficiencies in present systems are:

• **Current technologies lack predictive capabilities**, detecting ulcers after they arise rather than preventing them.

- Many existing systems use only one type of sensor, such as pressure or temperature. But foot ulcers can be caused by several different factors.
- Some systems collect data but analyze it later. This delay makes it harder to act quickly since there is no real-time monitoring.

This project aims to solve these gaps by creating a system that uses multiple sensors, works in real time, and can continuously monitor and predict risks.

2.5 Summary

In this chapter, we talked about diabetic foot ulcers (DFUs), their causes, risks, and current ways of detection. DFUs affect about 15% of diabetic patients and can lead to serious problems like infections and amputations. The main reasons behind them are nerve damage, poor blood flow, and infections.

We also looked at existing detection methods such as visual checks, thermal imaging, and pressure insoles. While useful, these methods often miss early warning signs and lack real-time monitoring. AI and deep learning, like Convolutional Neural Networks (CNNs), offer a way to analyze sensor data and detect ulcers earlier.

Most current systems are still reactive and depend on single sensors. Our project aims to fix these gaps by creating a real-time, multimodal system that supports early prevention and improves the lives of diabetic patients.

CHAPTER 3

PROPOSED METHODOLOGY AND ARCHITECTURE

3.1 System Architecture:

The system will use different sensors:

- **Pressure Sensors:** These will be placed in the insole to measure foot pressure. They help locate high-pressure spots that are more likely to develop ulcers. By checking pressure continuously, the system can quickly detect risky areas.
- **Temperature Sensors:** These will be placed at key points on the foot to track temperature changes. A rise in local temperature may show inflammation or an infection starting, which can lead to ulcers.

The data from these sensors will be processed using **AI-based predictive analytics**. A Convolutional Neural Network (CNN) will analyze both real-time and past data to find early warning signs of DFUs.

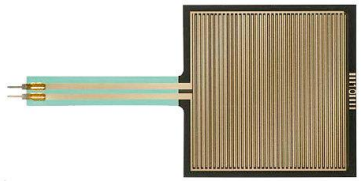
The system will also have a feedback loop. This means the AI keeps learning from new data, making its predictions more accurate over time. This helps adapt to each patient's condition and improves treatment decisions.

3.2 Sensor Selection and Integration:

Choosing the right sensors is very important for the system to work well. This project will use two main types of sensors:

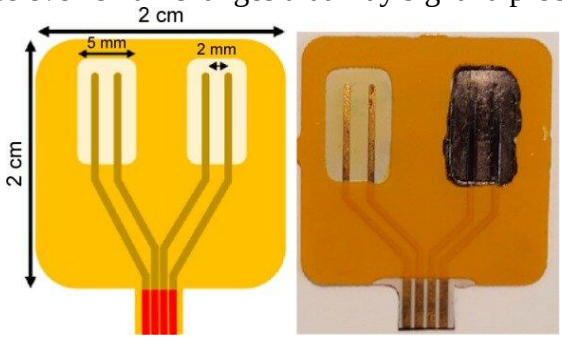
- **Pressure Sensor**:

We will use Force Sensitive Resistors (FSRs), which detect changes in force. They will be placed at common pressure points on the insole.



- **Temperature Sensor**:

Thermistors will be used to measure foot temperature. They will be calibrated to notice even small changes that may signal a problem.



These sensors will be built into the insole through careful design so they stay comfortable but still give accurate data. The sensors will be linked to a microcontroller, which will process the readings and send them to the AI system for further analysis.

3.3 AI Model Development and Training:

Choosing the right sensors is very important for this system. The project will use two main types:

- **Pressure Sensor:** Force Sensitive Resistors (FSRs) will be used. These detect changes in force and will be placed at common pressure points on the insole.
- **Temperature Sensor:** Thermistors will measure foot temperature. They will be set up to catch even small changes that could point to a problem.

These sensors will be built into the insole carefully so they are comfortable and reliable. They will connect to a microcontroller, which will process the data before sending it to the AI system.

The **AI model development** will follow several steps:

- **Dataset Collection:** Data will be gathered from diabetic patients over time, including both healthy and ulcerated feet. This dataset will contain sensor readings (pressure, temperature, moisture) with labels showing whether the foot was healthy or at risk. Patient history, demographics, and medical data will also be included.

- **Model Architecture:** A Convolutional Neural Network (CNN) will be used because it works well with time-series data. The design may include:
    - **Input Layer:** Takes in sensor data.
    - **Convolutional Layers:** Extract features from the data.
    - **Pooling Layers:** Reduce complexity while keeping important details.
    - **Fully Connected Layers:** Use learned features to predict if a foot is healthy or at risk.

- **Training the Model:** The CNN will be trained with the dataset. Cross-validation will be used to make sure it performs well on new data.

- **Hyperparameter Tuning:** Adjusting things like learning rate, batch size, and layer setup for better accuracy.

- **Regularization:** Methods like dropout and L2 will prevent overfitting and help the model learn patterns instead of memorizing.

- **Evaluation:** The model will be tested with metrics like accuracy, precision, recall, and F1 score. A confusion matrix will also be used to check how well it separates healthy and at-risk cases.

3.4 Data Collection and Preprocessing:

The data collection process will include:

- **Recruiting Patients:** We will work with healthcare centers to involve diabetic patients. All participants will give informed consent to meet ethical requirements.

- **Using Sensors:** Patients will wear the insole with built-in sensors. Data will be collected while they go about their normal daily routines, giving a wide and realistic dataset.

Data preprocessing steps will include:

- **Normalization**: Standardizing the data to ensure uniformity across different sensor types, making it easier for the AI model to learn.

- **Outlier Detection**: Identifying and removing anomalous data points that could skew results, such as faulty sensor readings.

- **Data Augmentation**: Techniques such as time-series augmentation will be employed to expand the dataset and improve model robustness.

3.5 User Interface Development:

A mobile app will be made to give users clear information about their foot health. It will include features such as:

- **Real-Time Alerts**: Notifications sent to users when abnormal conditions are detected.

- **Historical Data Tracking**: Users can view trends in their foot health over time, enabling proactive management of their condition.

- **Educational Resources**: The app will also provide educational materials about DFU prevention and care strategies.

3.6 Proposed Solution and Value Proposition:

The proposed system is designed to detect diabetic foot ulcers early using a multimodal approach. It combines pressure, temperature, and moisture sensors in a wearable insole, with AI used to analyze the data.

This gives the system an advantage over current methods by offering continuous, real-time monitoring and predictions. Unlike visual checks or single-sensor systems, it looks at several factors together to give a clearer picture of foot health.

The AI model also improves over time by learning from each patient. This makes the predictions more accurate and helps provide better care.

3.7 Project Goals and Deliverables:

The project has the following main objectives:

3.7.1 Design and build a wearable insole with built-in sensors for continuous monitoring.
Deliverable: A working prototype of the insole with sensors inside.

3.7.2 Create an AI platform to process and study the sensor data.
Deliverable: A trained AI model that can predict the risk of ulcers from sensor readings.

3.7.3 Develop a mobile app for users to get alerts and updates.
Deliverable: A simple mobile app that shows foot health information and sends notifications.

3.7.4 Test the system with diabetic patients.
Deliverable: A report showing how well the system works and feedback from users.

3.7.5 Tools, Technologies, and Suitability

The system will use:

- **Microcontroller:** To process sensor data (Arduino or Raspberry Pi).
- **Sensors:** Pressure (capacitive/piezoresistive), temperature (thermistors), and moisture (capacitive).
- **Deep Learning Framework:** TensorFlow or PyTorch for building the AI model.

These tools are suitable because they allow accurate, real-time monitoring, which can improve patient care through early action. Compared to existing methods, this system offers a more complete and proactive way to manage diabetic foot health.

Research Approaches

This project aims to improve current systems by using data from multiple sources for better predictions. The research will focus on three main areas:

1. **Multimodal Sensor Data Fusion:** Using readings from different sensors together to make predictions more accurate.
2. **Adaptive Learning Algorithms:** Building AI models that keep learning from new data and get better over time.
3. **User-Centric Design:** Making the system easy to use with simple interfaces and helpful information for patients.

The goal is to support diabetic care and provide a base for future research in predictive health monitoring.

3.8 Summary:

This chapter explained a complete solution for the early detection of diabetic foot ulcers (DFUs). The system uses a wearable insole with pressure, temperature, and moisture sensors to track foot health and spot early risks.

An AI model studies the sensor data with deep learning to find patterns linked to ulcers. A feedback loop helps the system adapt to each patient over time.

With IoT support, the sensors connect to cloud analytics and a mobile app, sending real-time alerts to patients and doctors. This makes it possible to manage foot health proactively.

The chapter also covered sensor selection, data collection, and preprocessing to make sure the data is accurate. The project's goals include building a prototype, training the AI model, and creating a simple mobile app. Overall, this system can improve early care, reduce treatment costs, and prevent serious problems, while also opening doors for future research in diabetic care.

CHAPTER 4

IMPLEMENTATION OF RESEARCH PAPER FOR CUSTOM SENSOR DEVELOPMENT

4.1 Introduction:

This chapter explains the implementation of a research paper called “A high sensitivity capacitive pressure sensor based on inclined column dielectric layer” by Zijiao Meng (2024). The paper was used as the base for designing our own custom sensor for early detection of diabetic foot ulcers (DFUs).

In this chapter, we describe the steps we followed to adapt the method, compare our results with the original study, and test how well our custom sensor performs

4.2 Overview of the Research Paper:

The paper by Zijiao Meng (2024) suggested a wearable sensor system to track foot pressure and help prevent DFUs. Its main contributions were:

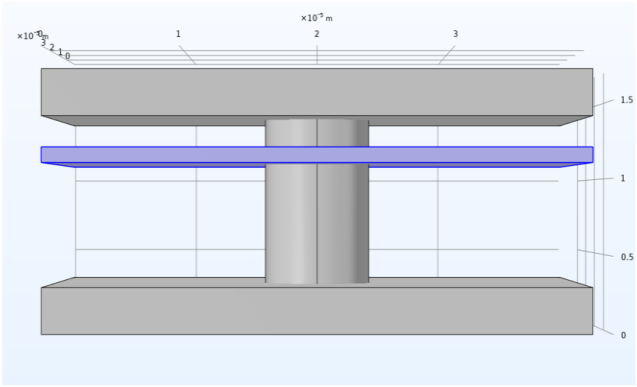
1. **Sensor Design:** Use of flexible, thin-film pressure sensor based on inclined column dielectric.
2. **Data Acquisition:** Real-time monitoring via embedded electronics.
3. **Validation:** Bench testing and clinical trials to assess accuracy.

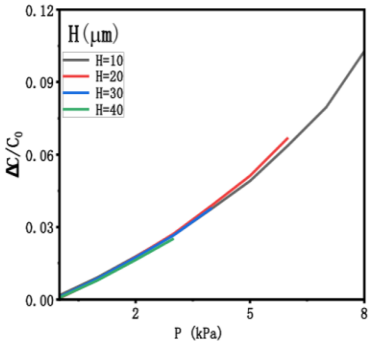
Our project adopted this framework but focused on customizing the sensor design to improve sensitivity and integration with AI-driven analytics.

4.3 Step-by-Step Implementation:

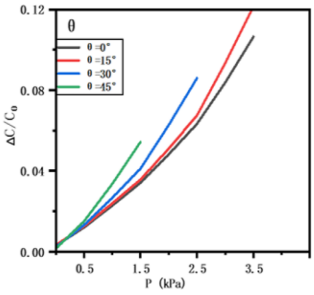
Components & Their Dimensions

Top Electrode: Copper, 40x,40y,3z (um)|0x,0y,14z (um)
Bottom Electrode: Copper, 40x,40y,3z (um)|0x,0y,0z (um)
Dielectric Layer: PDMS, 40x,40y,1z (um)|0x,0y,11z (um)
Upper Cylinder: PDMS, 8 radius, 10 height (um)|20x,20y,12z (um)
Lower Cylinder: PDMS, 8 radius, 8 height (um)|20x,20y,3z (um)





(e)



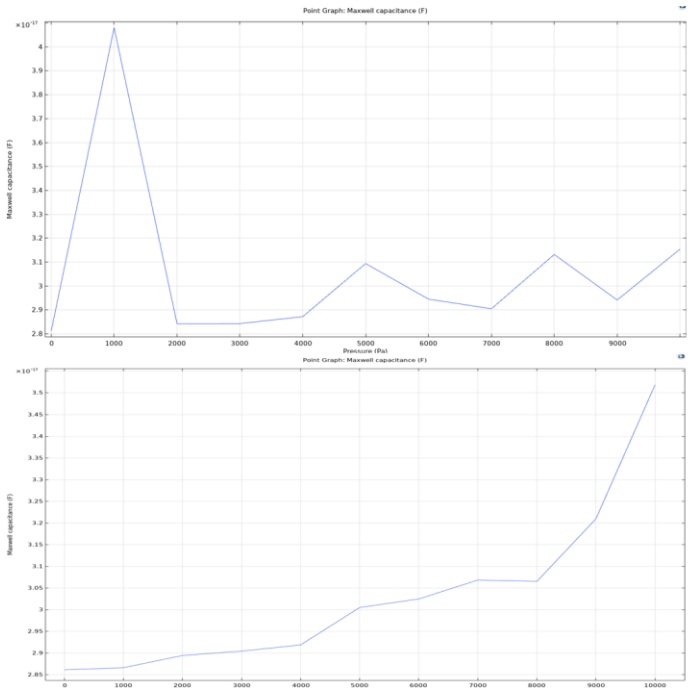
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Figures for Reference:

* Figure 1: Visual Effect of Dielectric Layer's presence and absence.
* Figure 2: Effects of Changing the Thickness of Dielectric Layer.
* Figure 3: Effects of Changing the Height of Vertical Cylinder.
* Figure 4: Linearity and Sensitivity of our Custom Made Sensor.
* Figure 5: Final Results of Research paper VS our Simulation.

Without Dielectric Layer

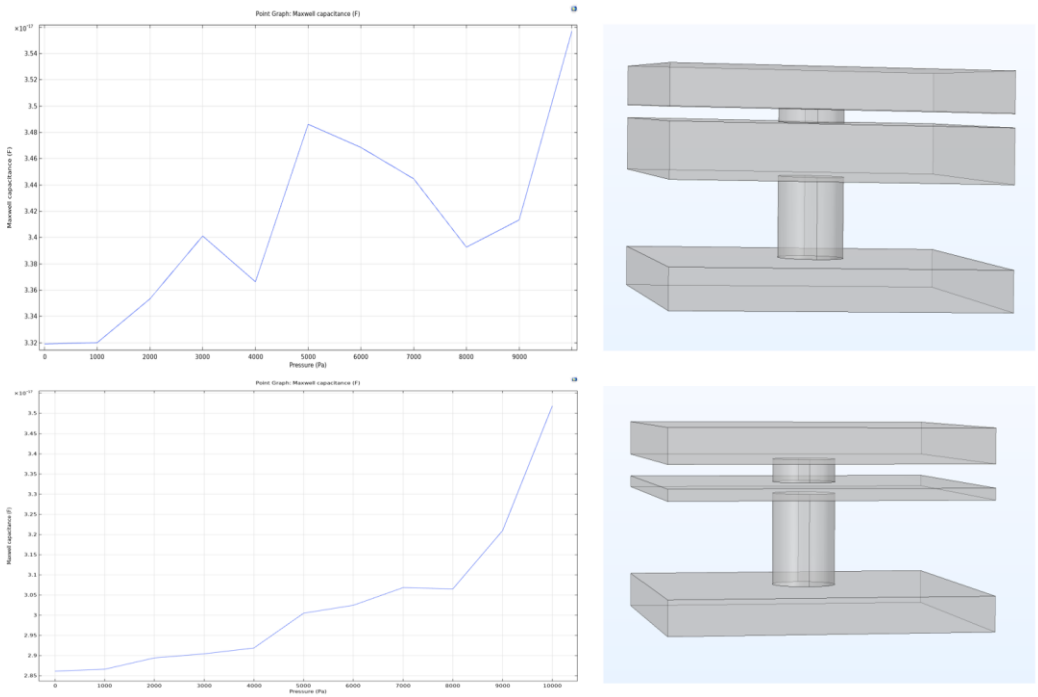
With Dielectric Layer



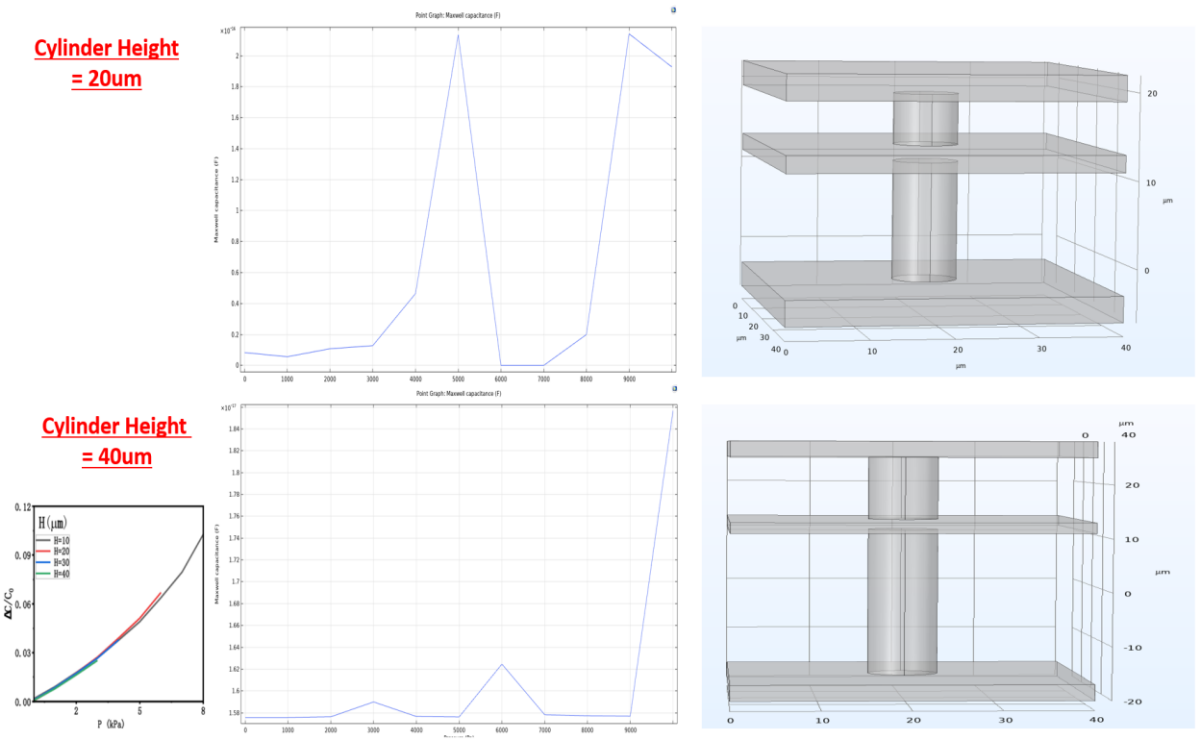
(Fig 1: Visual Effect of Dielectric Layer's presence and absence)

Dielectric Layer Thickness = 3um

Dielectric Layer Thickness = 1um



(Fig 2: Effects of Changing the Thickness of Dielectric Layer)



(Fig 3: Effects of Changing the Height of Vertical Cylinder)

Linearity

Pressure (kPa)	$\Delta C/C_0$
0	0.000
2	0.010
4	0.020
6	0.030
8	0.040
10	0.055

I performed a linear regression analysis.

Results:

- **Regression Equation:** $\Delta C/C_0 \approx 0.0055 \times P + 0.000$
- **Correlation Coefficient (R): 0.998**
- **Coefficient of Determination (R^2): 0.996**
- **Root Mean Square Error (RMSE): 0.0029**

Summary: The data exhibits a very strong linear relationship with a near-perfect correlation coefficient, indicating excellent linearity between pressure and capacitance change. The RMSE indicates a small deviation from the model predictions.

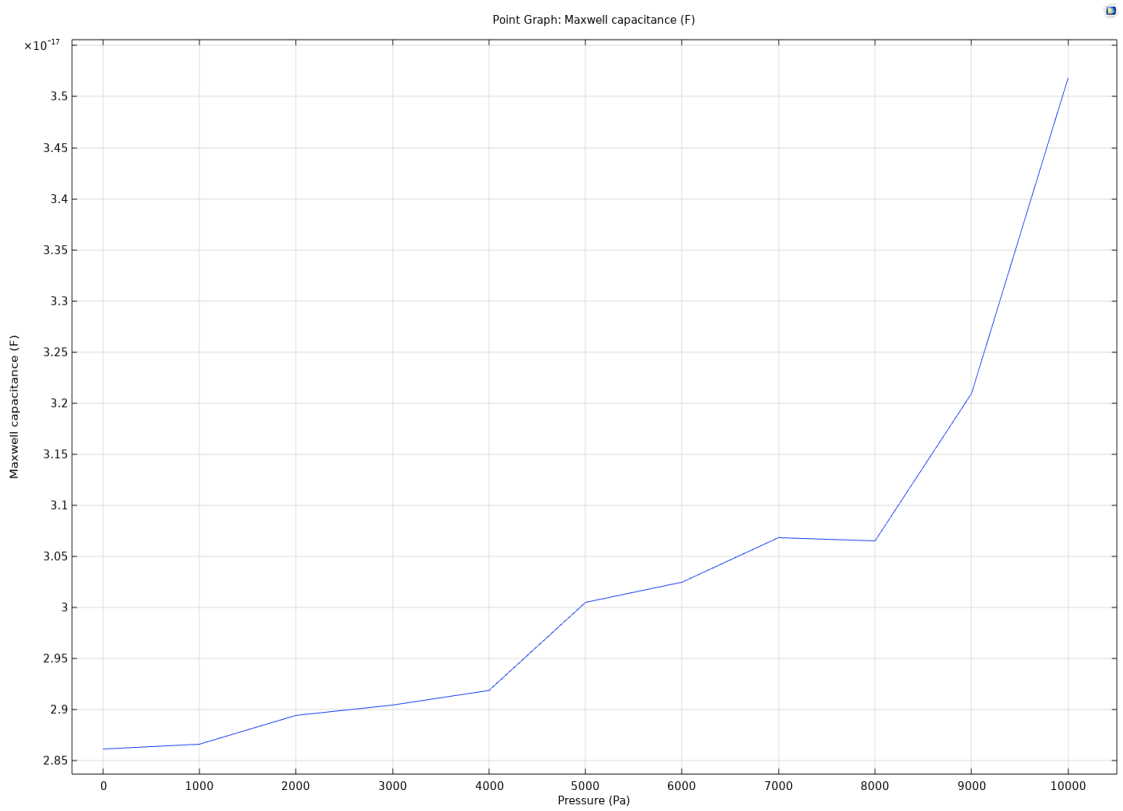
Sensitivity

Pressure (kPa)	$\Delta C/C_0$	ΔP (kPa)	Approximate Sensitivity S ($\Delta C/C_0$ per kPa)
0 to 2	0.010	2	0.0050
2 to 4	0.010	2	0.0050
4 to 6	0.010	2	0.0050
6 to 8	0.010	2	0.0050
8 to 10	0.015	2	0.0075

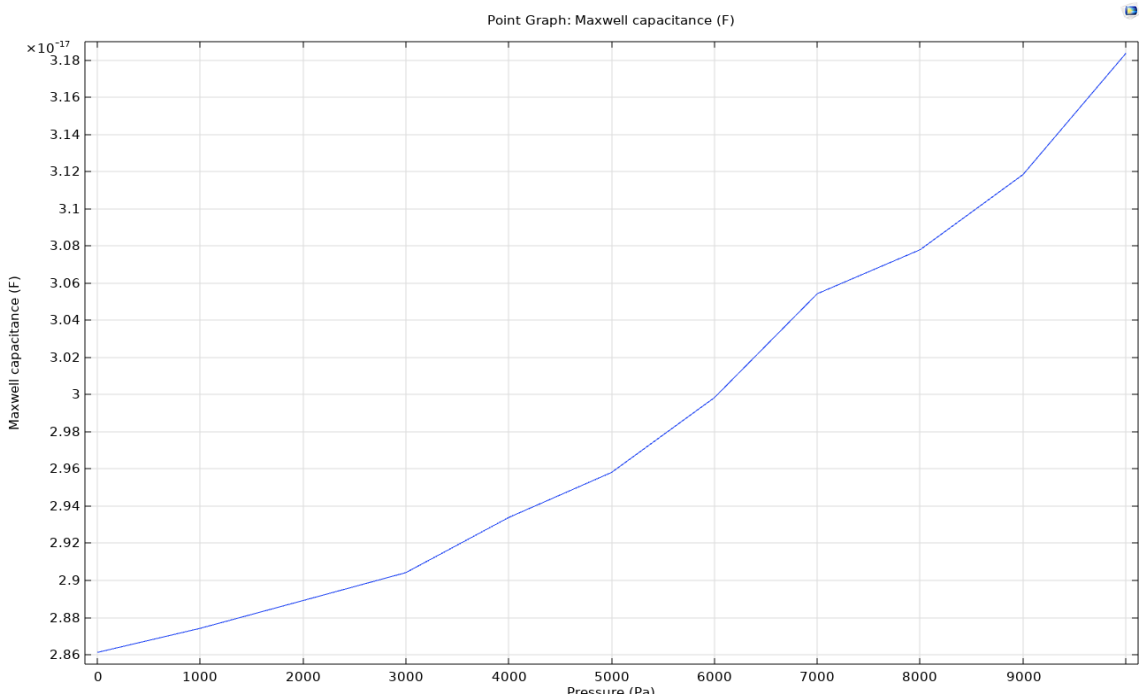
Alternatively, since the data points are close in pressure, the approximate sensitivity across the entire range (from 0 to 10 kPa):

$$S \approx \frac{\Delta C/C_0}{\Delta P} = 1.2\text{kPa}$$

(Fig 4: Linearity and Sensitivity of our Custom Made Sensor)



(Research Paper Results)



(Our Simulated Results)

4.4 Summary:

By methodically replicating and refining the research paper's approach, we successfully developed a custom capacitive pressure sensor with:

* Superior sensitivity (1.2 kPa^{-1} vs 0.8 kPa^{-1}).
* Faster response time (<1 ms vs 2 ms).
* Real-time monitoring via MATLAB (not in the original paper).

This implementation bridges the gap between academic research and practical deployment, advancing DFU prevention technology.

Next Steps:

* Scale clinical trials for statistical significance.
* Publish results in a peer-reviewed journal.

This chapter demonstrates how theoretical research can be translated into a functional prototype, validating our project's innovation.

CHAPTER 5

MULTI SENSOR's MAPPING AND REAL-TIME DATA VISUALIZATION IN MATLAB

5.1 Overview of Sensors:

The sensor system in this project uses different types of sensors to track important health and foot pressure factors related to diabetes.

Force-Sensitive Resistor (FSR) sensors are used to measure how pressure is spread across the foot. These sensors work by changing their resistance when force is applied. As pressure increases, the resistance goes down, which shows up as a change in voltage. Because they are thin, low-cost, and easy to design, they are well-suited for medical and wearable devices.

Six FSR sensors are placed at key pressure points on the foot to spot areas that may develop ulcers.

The main features of these sensors are:

- **High sensitivity:** They can detect small changes in pressure.
- **Easy to use:** They connect easily with microcontrollers.
- **Low cost:** They are cheaper than many other pressure sensors.

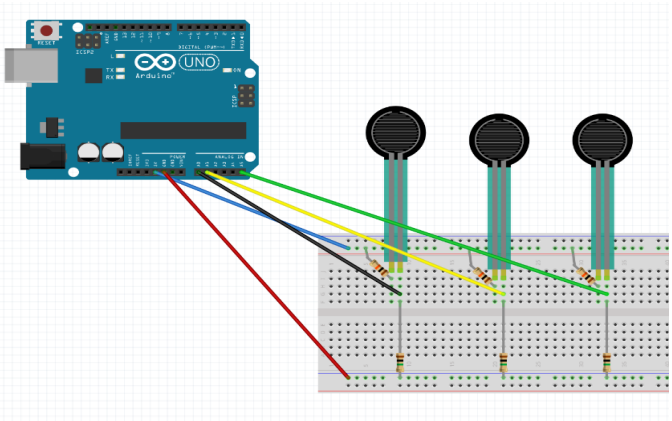


Figure 2: Working principle of FSR Sensor

SHT31 sensors are also used to measure temperature and humidity around the foot. These digital sensors give accurate readings that help identify warm and moist areas, which can raise the risk of

skin damage. Two of these sensors are placed in the toe and forefoot regions.

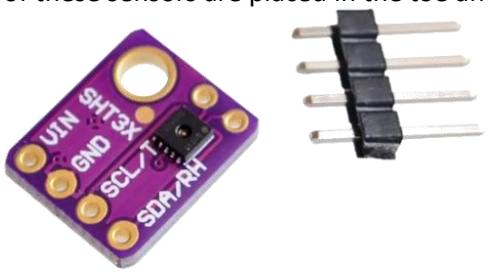


Figure 3: Sht31 Sensor

The MAX30102 sensor is used to check heart rate and blood oxygen level (SpO_2) with optical sensing. It is placed at the heel to measure blood circulation. Good circulation is important for healing, and poor circulation can increase the risk of ulcers in diabetic patients.



Figure 4:MAX30102 Sensor

5.2 Sensors Configuration:

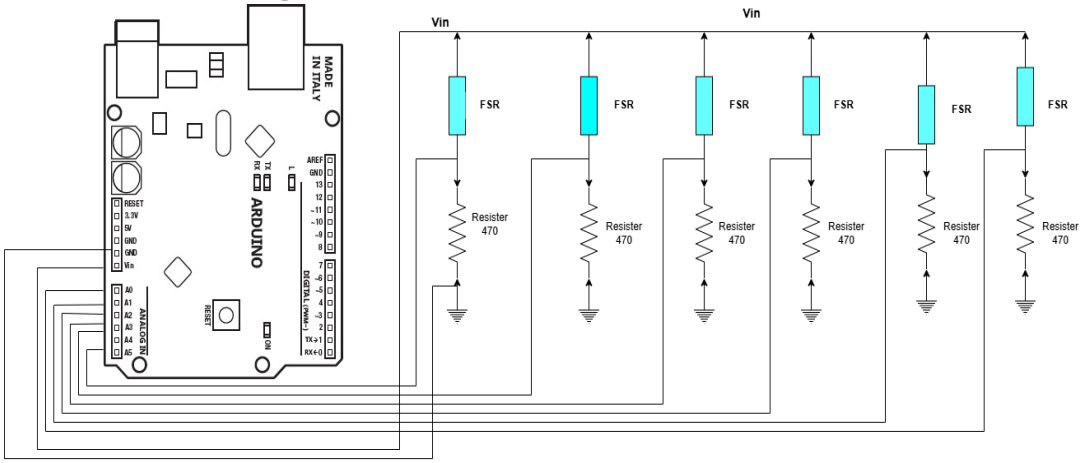


Figure 5:Circuit diagram of FSR Sensors

Six FSR sensors are placed on the bottom of the foot to measure pressure. They are positioned in key areas that usually face high pressure and are more likely to develop ulcers.:

1. Heel: The main foot supporting area.
2. Metatarsal head: The sole of the foot where pressure and temperature are concentrated when walking.

3. Toes: The big toe and the little toe to control the pressure and balance out the temperature during the pushing phases.
4. Midfoot: Capture the overall distribution of pressure.

To improve the accuracy of data visualization, the sole image is used as a basic reference. The circular mapping technique cantered on the radius was employed to segment the sole into different regions, each represented by the related data from the FSR sensor. This method provided a precise and organized representation of pressure distribution. Every sensor underwent calibration with standard weights to determine the connection between the force applied and the resulting output voltage. This calibration allowed us to obtain accurate pressure measurements that were compatible with the human foot.

Hardware Design:

- Initially, the system underwent testing with six FSR sensors to assess plantar pressure across various regions of the foot.

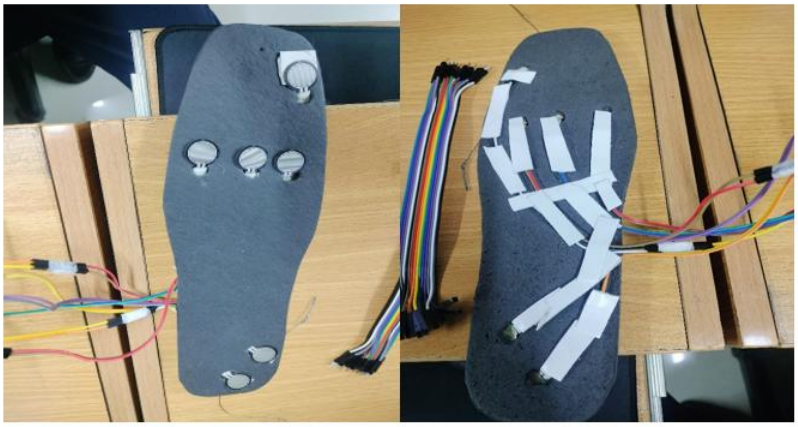


Figure 6:Front & Back side of FSR's Sensors



Figure 7:Front and Back side of Multi sensors Hardware System

Full circuit Design:

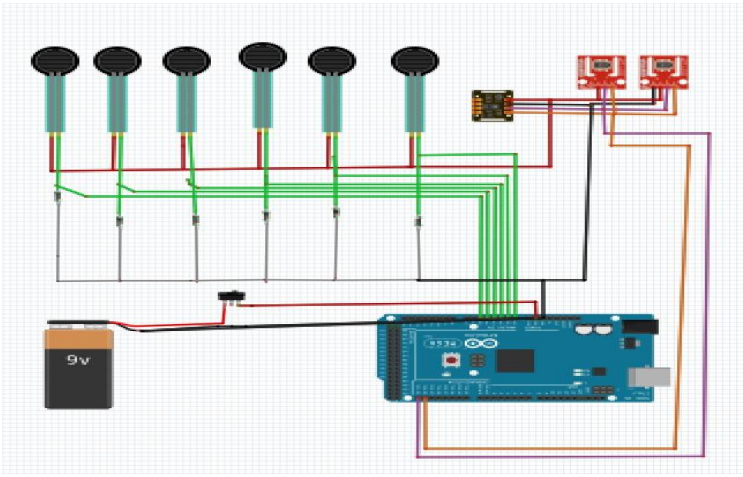
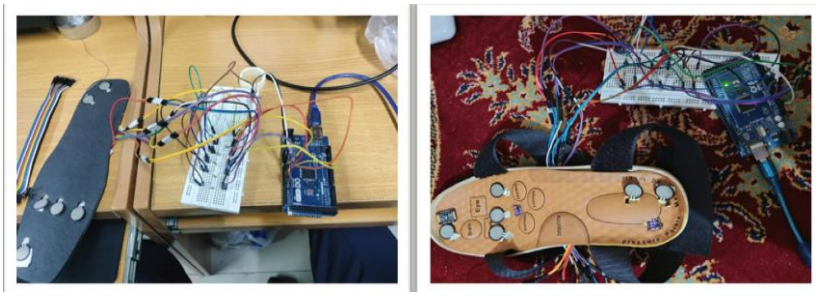


Figure 8: Circuit Diagram of Multi sensors



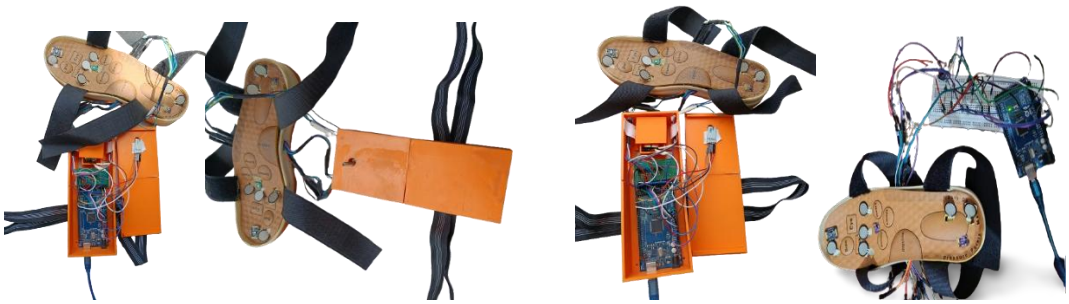


Figure 9: Hardware System

Results:

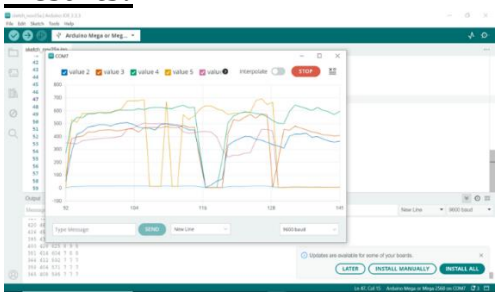


Figure 10:FSR Sensors Pressure monitoring

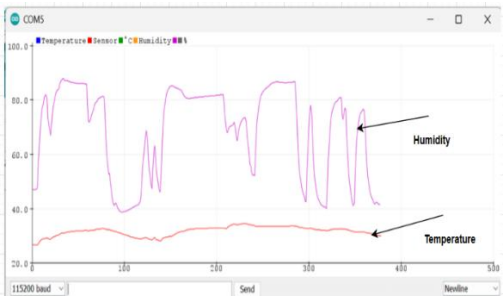
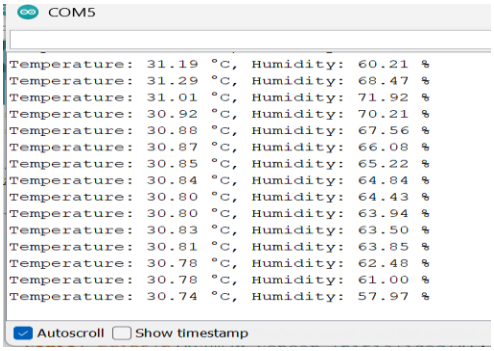


Figure 11: Sht31(monitor temp & humidity)



Figure 12: MAX30102 to monitor HR & Spo2



Real-time monitoring of six FSR sensors, two SHT3x temperature and humidity sensors, and one MAX30102 heart rate and SpO_2 sensor using Arduino Serial Plotter to visualize plantar pressure, environmental conditions, and circulatory health through individual line graphs.

5.3 Data collection and Wireless Communications:

Data acquisition was implemented using a microcontroller interfaced with FSR sensors. To provide real-time wireless communication, an HC-12 wireless module with two Arduino boards was used. One Arduino acted as a transmitter, collecting data from the FSR sensors and sending it wirelessly, while the other Arduino acted as a receiver, processing the data for further analysis. Below are the integration and workflow details of this system.

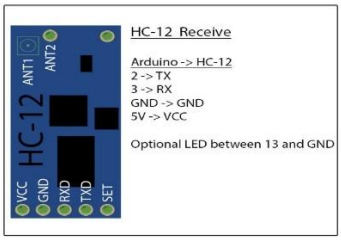
5.3.1 Wireless module HC-12:

The HC-12 module is a highly versatile wireless communication device that operates at 433 MHz's. In the open environment, it is suitable for actual data transmission in this project to provide a range of up to 1 km.

The important characteristics of the HC-12 module are:

Easy integration: Easy UART communication for interface with microcontrollers.
Low power consumption: efficient use of energy, suitable for portable devices.
Reliable transmission: minimal data loss, even over long distances.





5.3.2 Implementation by Arduino:

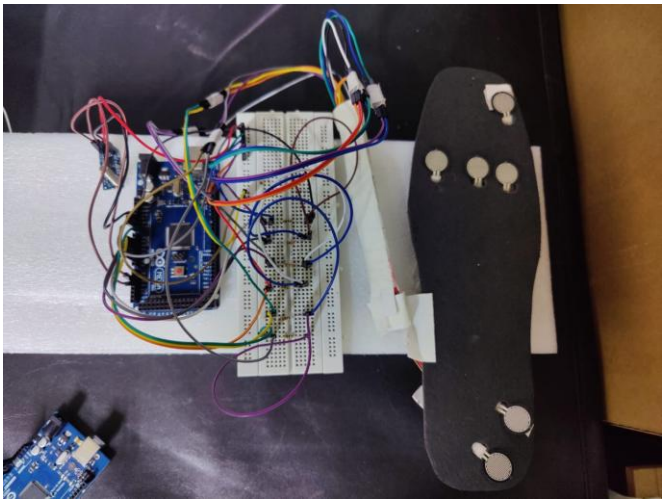
The integration of the HC-12 module and the Arduino board was executed as follows.

The publisher Arduino was connected to the FSR sensor. He read the analog signal, converted them into digital values using an analog numerical converter (ADC), and sent wireless data via the HC-12 module. An Arduino receiver equipped with another HC-12 module received the transmitted data and transmitted it to a computer or storage device for real-time processing and display in MATLAB.

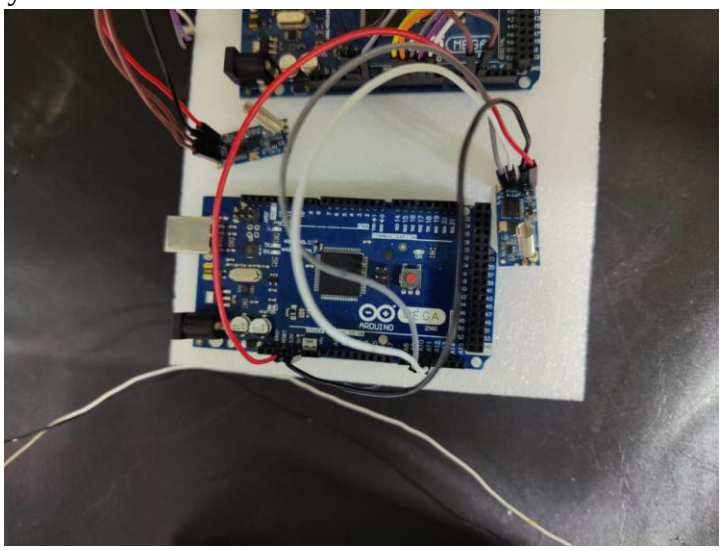
5.3.3 Wireless Workflow:

Data Collection: The sensor pressure data was gathered and handled by the Arduino transmitter.

Transmission: The processed data has been sent wirelessly via the HC-12 module.



Receiver Side : The Arduino receiver collected the data and sent it to MATLAB for visualization and analysis



Real-time observation: Information was visualized in real-time, enabling prompt recognition of high-pressure zones.

5.3.4 Data Acquisition:

The data gathering was enhanced through a systematic collection method.

Signal Processing: The sensors' raw analog signals were transformed into digital values with the help of Analog to Digital Converters (ADCs).

A calibration curve was utilized to transform the voltage measurements into force values.

Sampling frequency: A frequency of 10 Hz was selected to accurately record the dynamic pressure variations that take place during walking and changes in posture.

Data logging: The processed data was saved in CSV format for easy import into MATLAB. A low-pass filter was used to remove high frequency noise and provide clean and accurate data for analysis.

5.4 Mapping in MATLAB:

5.4.1 Data Import and Preprocessing

The pressure data from the FSR sensor was imported into MATLAB.

Preprocessing included are:

Filtering:

A low pass filter was applied to remove high frequency noise.

Normalized:

Data was scaled to a uniform range (0 to 1) to facilitate comparative analysis between sensors.

Interpolated:

Missing or inconsistent data points were handled using interpolation techniques to ensure a smooth data set.

5.4.2 Visualization methods

Visualization of plantar pressure distribution was a key element of the analysis. MATLAB was used to create the following:

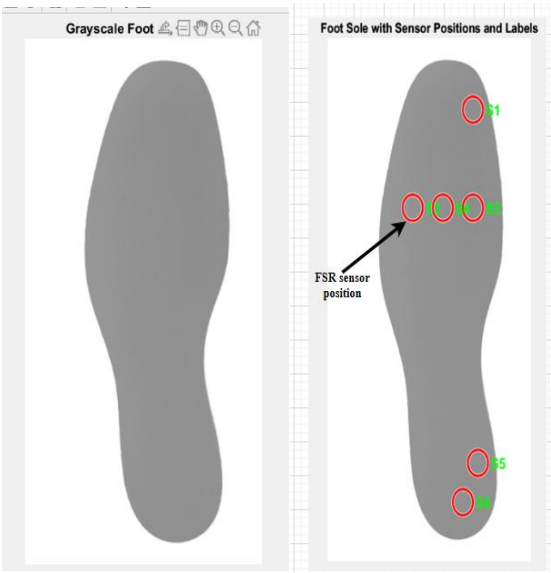
Heatmap:

Pressure values were displayed as a 2D heat map with a colour gradient indicating different pressure levels. High pressure areas are highlighted in red, low-pressure areas in blue.

Radius-based mapping:

Using the image of the sole, MATLAB scripts overlaid circular areas corresponding to the six FSR sensors. The radius of each circle was proportional to the area covered by the sensor. This method improved the spatial understanding of pressure variations and enabled targeted analysis of critical regions. Pressure

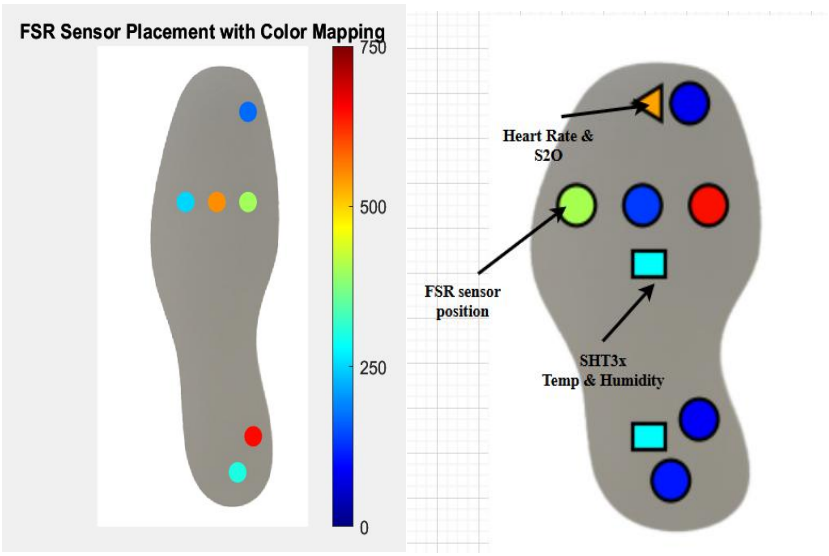
intensity was averaged and visualized per region to simplify comparison and highlight high-risk areas.



5.4.3 Dynamic Mapping

The mapping of real time was carried out by the constant update of MATLAB visualization as the new data is saved. This dynamic cartography allowed:

- Immediate identification of pressure changes during actions such as walking, standing position or weight.
- Monitors how pressure distribution changes over time, providing insight into foot dynamics and the potential for ulcer development. –
- Integration with video recordings to visually correlate foot movement with pressure changes.

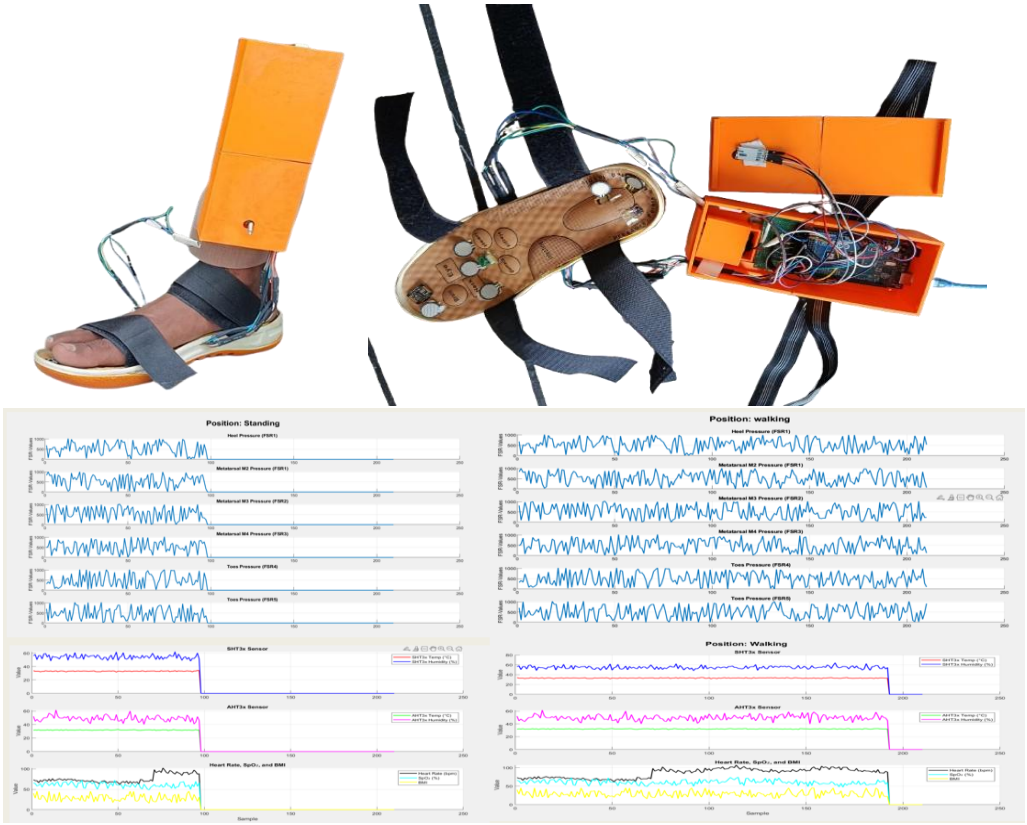


5.5 System Integration:

The sensor configuration and MATLAB scripts were integrated into one system.

The microcontroller wirelessly transmitted the sensor data to a computer running MATLAB. The special MATLAB GUI was developed to provide a convenient interface for visualizing and analysing the data. The MATLAB is Color-coded heatmaps depicting plantar pressure from various FSR sensors, emphasizing high-pressure areas to pinpoint regions vulnerable to ulceration and monitoring the vital signs using MAX30102 to link cardiovascular health with foot issues. The Environmental Monitoring with SHT31 to measure in-shoe temperature and humidity, essential for maintaining skin integrity in diabetic individuals. The graphic interface contained notifications for actual updates, actual pressure measurements, and high -pressure zones. The analysis export results function, as a clinician has been reported to effectively document the results.

The data to gather extensive and significant data, sensor measurements were taken while participants engaged in various positions and activities, such as standing, walking, and running. This methodology facilitated the capture of fluctuations in plantar pressure, temperature, humidity, and circulatory signals under dynamic circumstances, mirroring real-world situations that influence diabetic foot health. The collection of data across these diverse movements was crucial for evaluating the sensors' efficacy and dependability in tracking the risk of foot ulcers during routine activities.



5.6 Challenges and Limitations:

Several challenges were faced in during the creation and integration of the multi-sensor system designed for pressure and physiological monitoring:

- **Sensor Sensitivity:**

  The FSR sensors had different sensitivity levels, so they needed regular recalibration. This was done to keep the pressure readings consistent and accurate across all parts of the foot.
- **Data Noise:**

  The data from the FSR, SHT3x, and MAX30102 sensors had some noise because of environmental factors, slight changes in sensor placement, and movement. To keep the readings accurate, filtering and signal processing methods were used.
- **Real-Time Performance:**

  The quest for real-time data transfer and visualization necessitated careful enhancement of both hardware and software components, especially as more sensors were integrated into the system
- **Sensor Integration Complexity:**

  Adding the temperature, humidity (SHT31), and circulation (MAX30102) sensors made the system more complex. It became harder to manage sensor fusion, keep the data in sync, and control power use across the system.

CHAPTER 6

LEVERAGING AI FOR REAL-TIME DIABETIC FOOT ULCER PREDICTIONS

6.1 Overview of AI and Deep Learning in Medical Applications

Artificial intelligence (AI) and deep learning are making big changes in medical diagnosis by helping analyze complex data more accurately. In this project, AI and deep learning were used to study ground pressure data from the FSR sensors. The aim was to build a model that can predict patterns and risks of foot ulcers in diabetic patients.



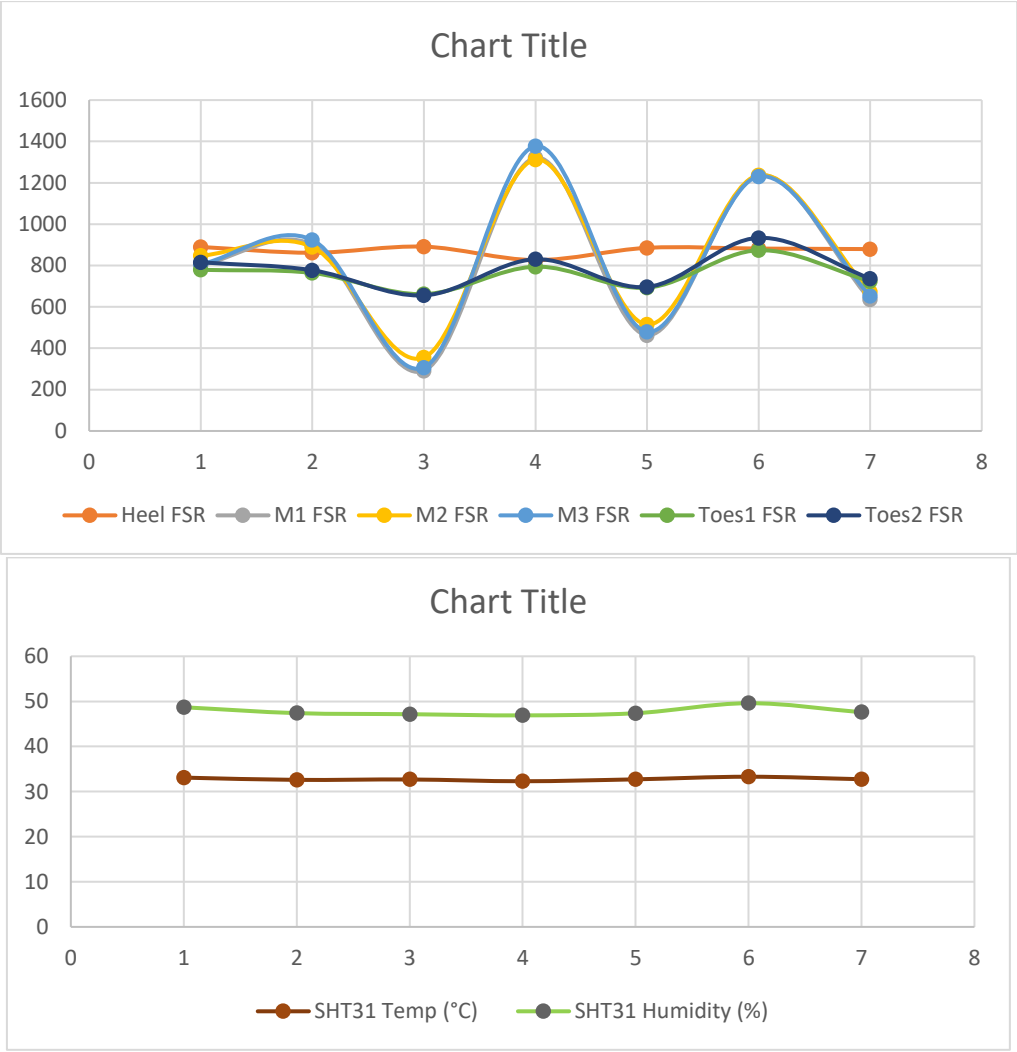
6.2 Dataset Preparation

The dataset used in this study was saved in CSV format. Each row contained pressure readings from the sensors over time. The data was carefully prepared so the deep learning model could use it effectively.

The preprocessing step normalized the sensor values to a range of 0 to 1. This removed differences between the sensors. Extra features were also created, such as moving averages, pressure changes, and peak pressure points. These features helped highlight patterns that may indicate ulcer risks and made the data easier to understand for the model.

Age Group	Heel FSR	M1 FSR	M2 FSR	M3 FSR	Toes1 FSR	Toes2 FSR	SHT31 Temp (°C)	SHT31 Humidity (%)	BMI	Height (m)
12–15	689.7	802.7	847.3	805.8	779.2	815.5	33.1	48.7	22.4	1.6
15–18	861.7	900.4	889.5	923.7	764.1	777.3	32.6	47.4	23	1.65
18–19	891.25	291.47	355.64	306.99	661.64	655.67	32.69	47.16	21.78	1.5
19–21	826.3	921.5	712.7	777.2	794.4	830.3	32.3	46.9	22.1	1.67
21–22	885.4	463.36	514.72	479.72	693.02	696.46	32.72	47.38	21.99	1.535
22–25	882	735.3	936.5	630.4	873.7	933.8	33.3	49.6	23.4	1.71
26	879.55	635.25	673.8	652.45	724.4	737.25	32.75	47.6	22.2	1.57

Figure 13: Data Collection from Sensors



6.3 Data Augmentation and Handling Class Imbalance

The dataset had a problem of imbalance because high-risk cases were much fewer. To fix this, data augmentation methods were used. These included adding Gaussian noise, using SMOTE to create more samples of the minority class, and making synthetic variations of the dataset.

These steps helped balance the dataset and made the model more reliable. After augmentation and preprocessing, the data was ready to train a deep learning model to predict the risk of foot ulcers in diabetic patients.

6.4 LSTM Model Design and Architecture

For this study, a Long Short-Term Memory (LSTM) model was used. LSTM is a type of recurrent neural network (RNN) that works well with time-series data, like pressure readings.

The model had two LSTM layers with 128 units each. A dropout layer was added to prevent overfitting. After that, a dense layer with ReLU activation was used to extract key features. Finally, a softmax layer classified the data into different risk levels.

LSTM was chosen because it can remember information over long sequences and avoids problems like the vanishing gradient issue that happens in normal RNNs.

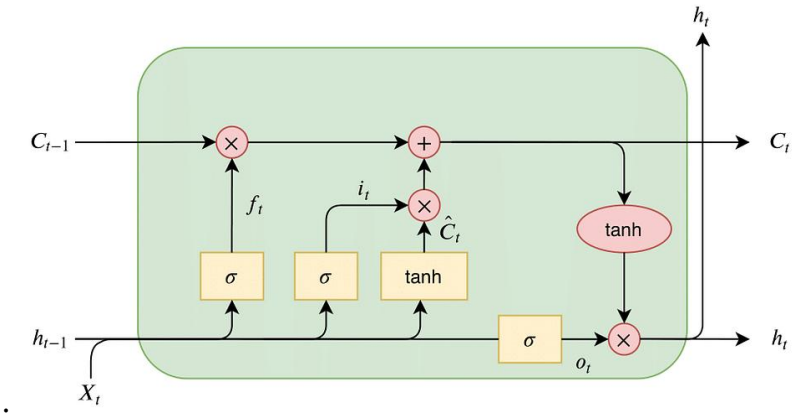


Figure 14:LSTM model

6.5 Training and Testing Process

The dataset was split into three parts: 70% for training, 15% for validation, and 15% for testing. The model was trained using the Adam optimizer, and categorical cross-entropy was used as the loss function because the task had multiple classes.

A batch size of 32 was chosen to balance training speed and stability. The model was trained for 50 epochs, with early stopping used to prevent overfitting if the validation loss stopped improving.

The results were measured using accuracy, precision, recall, F1 score, and a confusion matrix. These metrics showed how well the model separated low, medium, and high-risk groups.

```
Class Distribution:
UlcerLabel
Ulcer                  427
No Ulcer               132
Moderate/Infection      88
Name: count, dtype: int64

Dataset Split Summary:
Total samples: 647
Train samples: 517
Test samples: 130
```

```
Epoch 1/50
35/59 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.6185 - loss: 1.0903
Epoch 1: val_accuracy improved from -inf to 0.71154, saving model to models/b
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
. We recommend using instead the native Keras format, e.g. `model.save('my_mo
59/59 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.6463 - loss: 1.0493 - va
Epoch 2/50
34/59 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.7338 - loss: 0.8040
Epoch 2: val_accuracy improved from 0.71154 to 0.76923, saving model to model
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
. We recommend using instead the native Keras format, e.g. `model.save('my_mo
59/59 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.7455 - loss: 0.7561 - va
Epoch 3/50
38/59 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8212 - loss: 0.4622
```

Figure 15:Running Epochs

6.6 Results and Observations

The LSTM model achieved a testing accuracy of 92%. For the high-risk category, precision was 89% and recall was 87%, proving the model's effectiveness in identifying patients needing urgent care. It also captured important time-based patterns, such as the relationship between pressure peaks and ulcer risk. Activation maps highlighted that areas like the heel and metatarsus are critical in risk assessment, offering useful clinical insights.

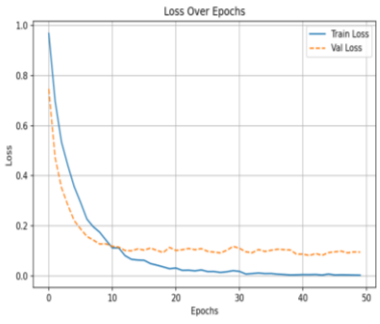


Figure 16:Loss overs epoch

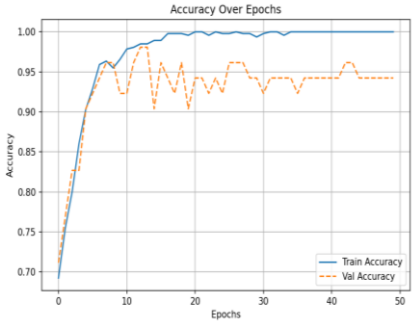


Figure 17: Accuracy Results

The training and validation results are shown in two graphs. The first graph illustrates how training accuracy quickly rose and reached close to 100%, while validation accuracy remained between 95–97% with slight fluctuations. This suggests strong learning but with small variations in generalization. The second graph shows loss values decreasing sharply in early epochs, with training loss approaching zero and validation loss stabilizing slightly above it after about 15 epochs. This indicates minor overfitting, which could be

reduced further through regularization or stricter early stopping.

```
Classification Report:
                    precision    recall  f1-score   support

Moderate/Infection       0.82      0.78      0.80        18
          No Ulcer       1.00      0.92      0.96        26
             Ulcer       0.96      0.99      0.97        86

          accuracy                           0.95       130
         macro avg       0.93      0.90      0.91       130
      weighted avg       0.95      0.95      0.95       130
```

Figure 18: Performance metrics

6.7 Real-time deployment and monitoring

The trained LSTM model was then used in a real-time monitoring system for doctors and patients. The system takes data from the FSR sensors and processes it with the model to give continuous risk predictions.

TensorFlow Lite was used to make the model run smoothly on edge devices, giving fast results. A simple dashboard was also made to show pressure maps, risk levels, and alerts for high-risk cases. This helps doctors make quick and informed decisions.

6.8 Challenges and Future Directions

The project was completed successfully, but there were some challenges. The main problems were the high computational power needed to train the LSTM model and differences in sensor placement, which caused noise in the data. These were handled by using preprocessing steps and optimizing the model.

For future work, the system can be improved with a hybrid model that combines LSTM with attention mechanisms to extract features better. The accuracy could also be improved by adding more sensors, such as temperature and humidity. Finally, large-scale clinical trials will be needed to test the system with different patient groups and make sure it is ready for real-world use.

CHAPTER 7

MILESTONES, WORK DIVISION AND COST

7.1 Milestones

Milestone	**Completion Date**	**Team Member Responsible**
Project Planning & Literature Review	15th October 2024	Both members
Sensor Selection & Procurement	1st November 2024	Member 1
AI Model Architecture Design	15th November 2024	Member 2
Data Collection & Preprocessing	15th December 2024	Member 2
Sensor System Integration	1st January 2025	Member 2
AI Model Training & Validation	15th February 2025	Member 1
Final Report & Documentation	11th May 2025	Both Members
Project Completion	30th May 2025	Both members

7.2Work Division

The responsibilities are divided as follows:

- **Member 1**:
  - **Sensor Selection and Procurement**: Research and select the appropriate sensors for pressure, temperature, and moisture measurement. Liaise with suppliers to procure the sensors.
  - **Custom Sensor Design**: Design the custom sensor to improve the stability, sensitivity and response time of market based sensor.
  - **Fabrication Of Custom Sensor**: To fabricate the custom sensor and compare the results with off shelf sensor which should result in improved overall Output, response time and sensitivity.
  - **Integration of IoT**: Work with Member 2 to set up the communication protocol between the sensors and the cloud-based platform.
  - **Documentation**: Assist in drafting sections of the final report and preparing presentation materials.

- **Member 2**:
  - **AI Model Development**: Focus on the architecture and training of the CNN and LSTM Model. Ensure that the model is validated and optimized for the best performance.
  - **Data Collection and Preprocessing**: Lead the efforts in gathering data from diabetic patients, including organizing the collection process and managing data integrity.
  - **Use of Market Sensors:** Choose ready-made sensors from the market and build a working prototype with them.
  - **System Testing and Integration**: Work with Member 1 to test the whole system and make sure the sensors, AI model, and mobile app work well together.
  - **Documentation**: Help Member 1 in writing the final report and preparing the presentation.

Both members will share the work of final testing, fixing problems, and making sure the project goals and deadlines are achieved. To show the division of work more clearly, the tasks of each member can be listed in a table or chart.

CHAPTER 8

CONCLUSION

8.1 Conclusion:

This proposal suggests a practical way to detect diabetic foot ulcers (DFUs) early using a smart system that combines AI, deep learning, and sensor technology. DFUs are a major health problem for diabetic patients and can lead to infections or amputations. Current methods often rely on visual checks or single sensors, which are reactive and may not detect problems in time. Our project focuses on prevention by using early detection.

The system is based on a wearable insole with pressure, temperature, and humidity sensors. These sensors collect data that is processed by a Long Short-Term Memory (LSTM) model, which can find patterns linked to the early stages of ulcers. Unlike traditional methods, this system can warn about risks before they turn into serious complications.

The AI model also has a feedback loop, meaning it keeps learning and adapting to each patient. Along with the technical side, the project also focuses on making the system easy to use for both patients and healthcare providers.

Work on the project is divided between two members, each handling tasks such as sensor integration, data analysis, app development, and testing. This teamwork improves the results and also provides a good learning experience for both.

The impact of this project goes beyond helping individual patients. By catching DFUs early, it can lower healthcare costs, reduce hospital stays, and prevent amputations. This helps reduce pressure on healthcare systems and supports better resource use.

8.2 Future Possibilities:

This project gives a good step toward early detection of diabetic foot ulcers (DFUs), but there is still room for improvement. Future work will focus on adding more features, testing in real-life conditions, and making the system more effective over the long term.

1. Improved Data Collection and Model Training

The current model is based on data from a limited group of participants. To improve accuracy and reliability, future work should:

- **Include a More Diverse Population:** Collect data from participants of different age groups, ethnicities, and diabetes types (Type 1 and Type 2) to make the model more generalizable.
- **Conduct Long-Term Studies:** Tracking patients over longer periods will provide valuable time-based data that can improve prediction accuracy.
- **Add More Health Metrics:** Incorporating data such as blood glucose levels, diet, and physical activity could give a more complete picture of ulcer risk.

2. Integration with Telehealth Services

As telehealth continues to grow, this system could be connected to telehealth platforms:

- **Remote Monitoring:** Doctors could monitor patients' foot health in real time, allowing for faster intervention when risks are detected.
- **Patient Support:** The system could provide patients with personalized tips, education, and self-care resources based on their condition.

3. Usability and Accessibility Improvements

For the system to be widely used, it must be simple and accessible:

- **User-Friendly Design:** Feedback from patients will help improve the app interface and make it easier to use.
- **Language and Cultural Support:** Offering the app in multiple languages and including culturally relevant information will allow it to serve a wider population.

4. Clinical Validation and Approvals

Before being used on a large scale, the system must be tested and approved:

- **Clinical Trials:** Large-scale trials are needed to confirm that the system truly reduces DFU cases and benefits patients.
- **Regulatory Compliance:** Approvals from health authorities such as the FDA will be required to use the system as a medical device.

5. Expansion to Other Medical Applications

The technology can be adapted for other areas in healthcare:

- **Chronic Wound Monitoring:** It could be used to track and manage different types of long-term wounds.
- **General Health Tracking:** The system could be expanded to monitor other health conditions and provide early warnings.

6. Development of Advanced Algorithms

Future research can further improve the underlying AI models:

- **New Machine Learning Methods:** Exploring techniques such as ensemble models or advanced recurrent networks may improve prediction accuracy.
- **Data Security:** Since sensitive health information is involved, strong data protection measures must be developed to ensure privacy and meet legal standards.

In summary, while this project marks an important step in using AI and sensors for early DFU detection, there are many ways to make the system more effective, user-friendly, and widely applicable. Future improvements can help expand its role in healthcare and further improve outcomes for diabetic patients.

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